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Prof. Kenneth M. Merz, Jr., Editor-in-Chief

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Dear Prof. Merz,

We submit our revised manuscript **“Computational Discovery of Antimalarial Candidates from African Natural Product Chemical Space”** for consideration as an Article in *JCIM* (Manuscript ID: ci-2026-012015). This work addresses a critical computational bottleneck in antimalarial drug discovery: conventional virtual screening approaches require screening tens of thousands of compounds per target (exceeding 1000 CPU-hours), making comprehensive multi-target exploration prohibitively expensive for research groups in malaria-endemic regions where 94 % of global cases occur.

Key innovations and significance:

Rigorously validated computational framework with dual-benchmark strategy. We developed a centroid-based screening protocol that achieves 99.3 % cost reduction (from 263,424 to 1,936 docking runs) while maintaining pharmaceutical-grade predictive accuracy. Critically, we established reliability through complementary validation strategies: (1) prospective DEKOIS external validation (PfDHFR, 40 actives + 1200 property-matched decoys) demonstrating that single-method AutoDock Vina docking yields near-random enrichment (ROC-AUC 0.450), and (2) retrodictive consistency against the MMV Malaria Box positive-control benchmark (399 confirmed antimalarial actives) where our dual-consensus protocol (Vina + ML-based DiffDock rescoring) achieves ROC-AUC 0.924–1.000 with bootstrap 95% confidence intervals across four *P. falciparum* resistance targets. This dual-benchmark approach—combining prospective external decoy discrimination with retrodictive in-distribution consistency—establishes predictive validity before extrapolation to unexplored African natural product space.

Generative scaffold-hopping beyond local neighborhood interpolation. Our ConvVAE-SMILES architecture systematically explores genuinely novel chemical space: 92.6 % of generated molecules are unreachable by local SELFIES perturbations from known antimalarials (mean max Tanimoto 0.206 to seed compounds), yet scaffold recovery reaches 69.3 % (70/101 bioactive frameworks preserved). The apparent paradox is resolved through topological analysis: the $1.84\times$ scaffold-to-whole-molecule Tanimoto ratio demonstrates systematic preservation of privileged ring systems while diversifying peripheral substituents. STONED-SELFIES validation confirms this result: 97.9 % of

VAE-generated molecules remain inaccessible even after 5,525 local mutations from known drugs, empirically demonstrating scaffold-hopping rather than mere interpolation.

Systematic characterization of African natural product chemical space. From 396 African natural products and 454 synthetic antimalarials, we generated a 65,856-molecule hybrid library exhibiting 31.4% scaffold diversity (20,702 unique Bemis–Murcko frameworks) with favorable drug-likeness (94.1% Lipinski-compliant, 57.0% SYBA > 0 for synthetic accessibility). Variational autoencoder clustering with KMeans identified 484 diverse centroids for efficient multi-target screening against PfDHFR, PfCRT, PfATP4, and PfClpP. The dual-filter consensus protocol (AutoDock Vina + DiffDock) exploits the orthogonality between physics-based and ML-based scoring (Spearman ρ = 0.1–0.4), reducing false positives by 12–45% relative to single-method approaches while recovering 69.8% of MMV Malaria Box actives at consensus thresholds calibrated against ChEMBL antimalarial binding affinities.

Computationally prioritized synthesizable leads with open-access dissemination. Multi-parameter optimization integrating binding affinity, geometric confidence, drug-likeness (QED), ADMET safety, and Lipinski compliance identified 19,913 synthesizable leads (SYBA > 0) from 53 optimization-worthy clusters. All 810 screened seed molecules with valid predictions exhibited predicted selectivity index > 10, and the top-20 MPO-ranked candidates share tractable piperazine scaffolds (SA < 2.0) with computationally predicted retrosynthetic routes dominated by reductive amination chemistry. Importantly, we quantified the orthogonality between drug-likeness and binding affinity through external Tartarus benchmark validation (19,913 molecules, Spearman ρ = 0.013, p = 0.091), confirming that MPO-based ranking captures developability criteria independent of raw docking scores. The complete computational workflow, validation protocols, and full dataset (including molecular structures, docking results, ADMET profiles, and retrosynthetic predictions) are deposited as an open-access community resource (Zenodo DOI: 10.5281/zenodo.19608875; GitHub: https://github.com/Vital-Sao/Malaria_codes).

Democratizing computational drug discovery in endemic regions. By reducing per-target screening requirements from >1000 CPU-hours to <10 CPU-hours, this framework enables comprehensive multi-target antimalarial screening on standard computational infrastructure. This accessibility gain is particularly critical for research institutions in malaria-endemic regions of sub-Saharan Africa, where 94% of global malaria burden occurs but computational resources for drug discovery remain severely limited. The validated efficiency-accuracy tradeoff establishes that resource-constrained settings need not compromise predictive quality, positioning local researchers to systematically explore regional chemical biodiversity for therapeutic candidates.

Relevance to *JCIM* readership: This work directly addresses core cheminformatics challenges: (1) generative molecular design with explicit validation of scaffold-hopping versus interpolation, (2) consensus scoring strategies for orthogonal computational evidence integration, (3) benchmarking protocols combining prospective external validation with retrodictive consistency checks, (4) multi-parameter optimization frameworks balancing potency and developability, and (5) computational efficiency innovations enabling drug discovery in resource-limited settings. The rigorous validation, open-source implementation, and complete data dissemination align with *JCIM*’s

emphasis on reproducible computational methodologies for chemical biology and drug discovery applications.

Suggested reviewers:

1. Dr. Maria Sorokina (University of Jena, Germany) — Expert in natural product cheminformatics and computational analysis of African medicinal plant databases; has published extensively on NP-like chemical space characterization and database curation methodologies.
2. Dr. Fidele Ntie-Kang (University of Buea, Cameroon) — Leading researcher in African natural product virtual screening and *in silico* antimalarial drug discovery; founder of the African Natural Products Database initiative.
3. Dr. David Rogers (Givaudan, USA) — Developer of extended-connectivity fingerprints and molecular similarity metrics; expert in fingerprint-based virtual screening and molecular diversity assessment.

This manuscript represents original work not under consideration elsewhere. All authors have reviewed and approved the submission.

Sincerely,

Myke Vital Sao Temgoua

(on behalf of all co-authors)